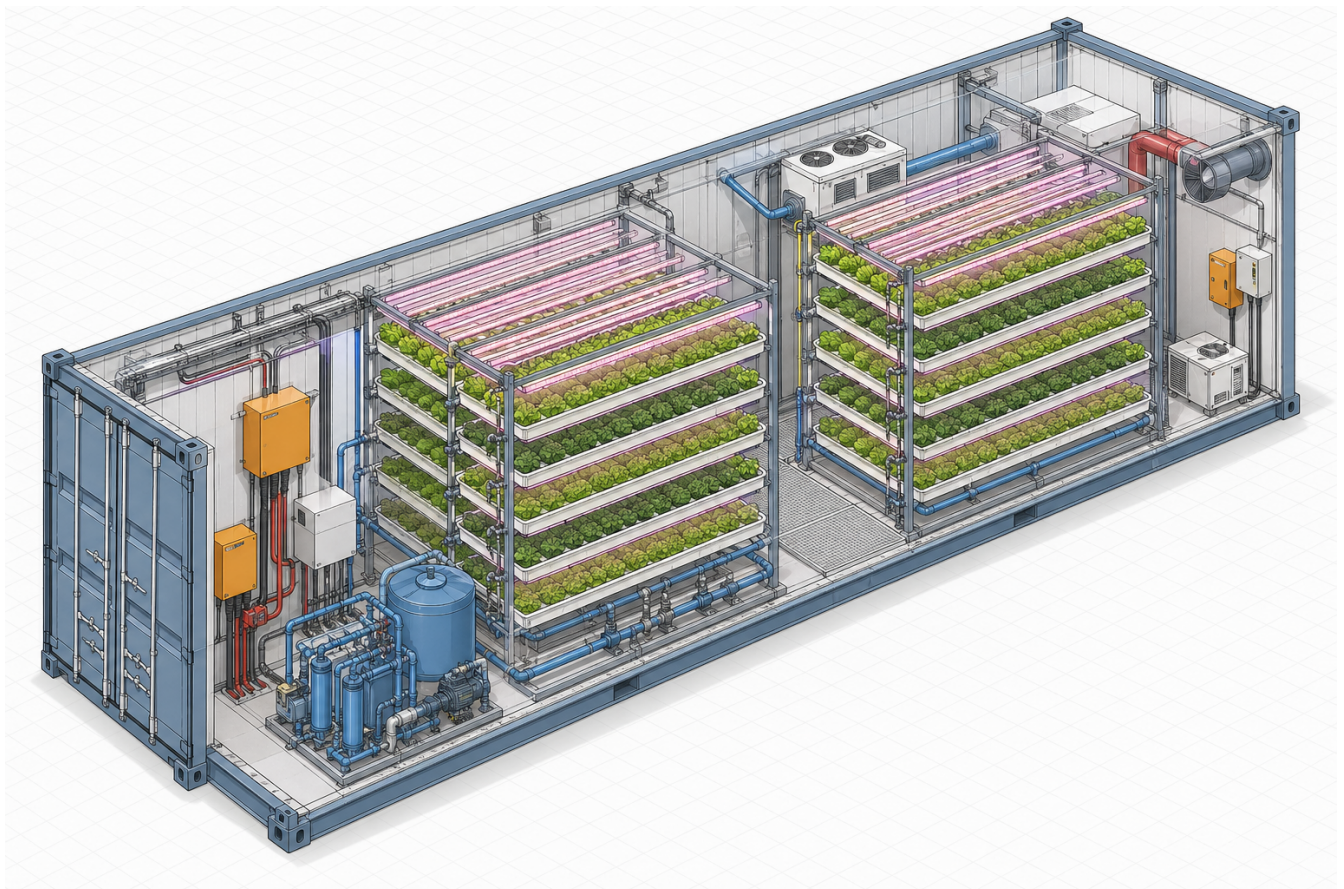


Modular vertical farm — 40ft container, leafy greens

Revision A · Fractional Forge · In build

Forge project pack

Project name	Modular vertical farm — 40ft container, leafy greens
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Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

TOTALS AT A GLANCE

MODULES 8	KEY PARTS 55	BOM ROWS 70	SUPPLIERS 32
FAILURE MODES 32	OPEN QUESTIONS 24	STANDARDS 6	REVIEWS 0
UNIT COST £69,555	CEILING £150,000	HEADROOM £80,445	

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1. Brief

FOUNDER SUBJECT

A 40-foot containerised modular vertical farm growing year-round leafy greens (lettuce, herbs, brassicas) for the United Kingdom market. Six thousand simultaneous plant slots across multi-tier hydroponic stacks, target yield four tonnes per year per container. LED grow-lights with daily light integral control, climate and carbon-dioxide dosing, recirculating nutrient system, IoT sensor mesh feeding a remote farm-management dashboard. Energy use under 250 kilowatt-hours per day in standard mode. Target installed capital cost under 150,000 pounds. Designed to ship complete on a flatbed lorry, plug into a three-phase mains supply, and run unattended between weekly plantings.

MISSION

Deliver year-round, unattended production of leafy greens inside a single 40-foot ISO container, giving UK food growers and retailers a turnkey, plug-in crop unit that ships complete on a flatbed lorry and yields four tonnes per year.

USE CASE

A food operator plugs a delivered container into a three-phase mains supply, loads seedlings weekly, and harvests lettuce, herbs, and brassicas continuously without on-site technical staff.

TARGET CUSTOMERS

UK supermarket supply-chain operators, farm-to-table catering groups, urban food hubs, local authority food-resilience programmes

WHY NOW

UK post-Brexit import friction and fresh-produce price volatility are pushing buyers toward domestic, controlled-environment supply; falling LED and sensor costs now make sub-£150,000 containerised yield economically viable at scale.

Constraints declared

Unit cost ceiling	£150,000
Max mass	5000.00 kg
Target process	not declared
Target material	not declared
Tolerance target	not declared
Quantity target	not declared

2. Regulatory posture

ISO 668

not-started

Series 1 freight containers — classification, dimensions and ratings

Defines the non-negotiable external envelope of the 40ft High Cube container shell used as the product chassis.

ISO 1496-1

not-started

Series 1 freight containers — specification and testing: general cargo containers

Structural and testing requirements for the container shell, including floor load and stacking loads.

IEC 60364

not-started

Low-voltage electrical installations

Governs the three-phase mains distribution, wiring, and protection devices installed inside the container.

UK Supply of Machinery (Safety) Regulations 2008

not-started

Machinery safety (UK retained law equivalent of Machinery Directive 2006/42/EC)

Applies to the automated nutrient dosing, HVAC, and moving parts within the container as an integrated machine.

UK Food Safety Act 1990 / EC 852/2004 (retained)

not-started

Food hygiene and food-contact materials

Interior lining, trays, and plumbing in contact with edible crops must meet food-contact and hygiene standards.

EA/Environment Agency EPR

not-started

Environmental Permitting Regulations — nutrient discharge

Recirculating nutrient effluent disposal may require an environmental permit depending on installation site.

3. Sizing optimisation

Forge ran a 35-trial sweep to find the best fit for this envelope. Coefficient library: vertical_farm v1.1.0.

Envelope

Warehouse bay (100 m²) · interior 10000×10000×6000mm · 100.00 m² floor

Final config

tiers: 4 · canopy_m2: 200 **[FEASIBLE]**

Founder's requested 200 m² × 4 tiers didn't fit; nearest feasible is 200 m² × 5 tiers (8.3 m² headroom).

Trial grid (rows = tiers, columns = canopy m² · cell = floor util %)

tiers “	100	150	180	200	230	260	300
2	89%						
3	65%	97%					
4	53%	78%	94%				
5	45%	67%	81%	90%			
6	40%	60%	72%	80%	92%		

Green = comfortable fit · Yellow = tight (e90% floor used) · Red = infeasible · Orange outline = winning config.

Top alternatives

tiers: 2 · canopy_m2: 100

-100 m² canopy vs winner, 2 tiers, 89.2% floor used · headroom 8.6 m²

tiers: 3 · canopy_m2: 150

-50 m² canopy vs winner, 3 tiers, 96.8% floor used · headroom 2.6 m²

tiers: 4 · canopy_m2: 180

-20 m² canopy vs winner, 4 tiers, 93.9% floor used · headroom 4.8 m²

Levers — what you could do next

Add one more tier (5 ' 6)

Gain: Canopy could grow to ~200 m² at same floor footprint

Cost: Rack height climbs to 3200mm — 2800mm ceiling headroom

Stretch to 230 m² canopy at 5 tiers

Gain: ~6000 g/wk more yield

Cost: Infeasible with current envelope: Floor over by 2.4 m² (3%)

Upgrade to a 53ft high-cube container

Gain: Adds ~20% floor area, ~300mm extra interior height — unlocks more tiers + more canopy in a single unit

Cost: Container cost rises ~30% (roughly +£3,000 CAPEX)

Engineering notes

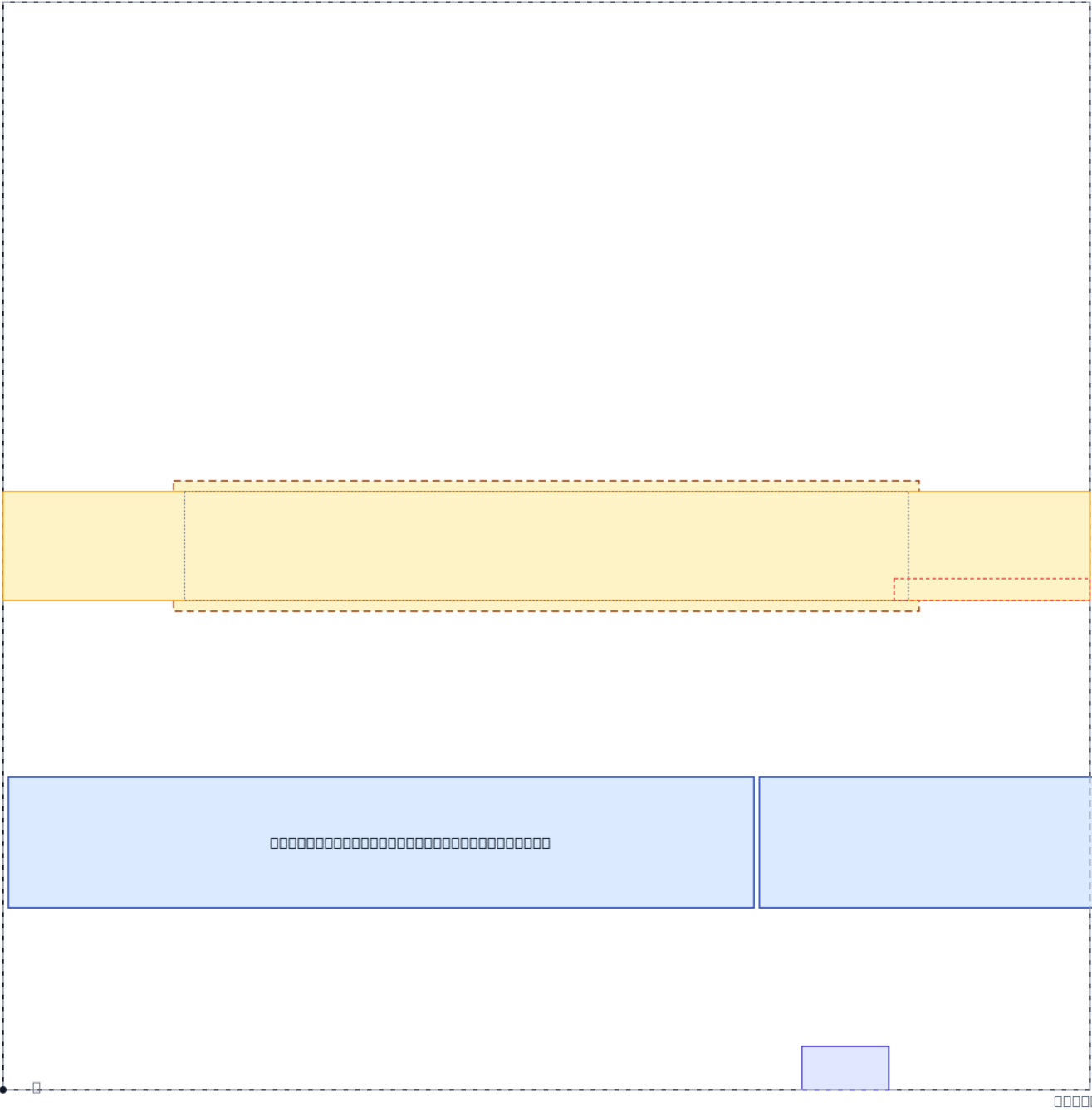
- Canopy density: 5 tiers × 85% aisle-efficient packing.
- Lighting installed: 250 W/m² (leafy greens at ~200 DLI).
- Transpiration load: 500 L/day water evaporates from canopy — ALL of it must be condensed by the HVAC evaporator coil and drained out of the envelope or humidity spikes and powdery mildew appears within days.
- Peak cooling budget: 30.0 kW sensible (LED heat) + 29.4 kW latent (condensing transpiration) = 59.4 kW total during photoperiod.
- Required dehumidification rate: 46.9 L/hour. Size the dehumidifier pint-rating at e 57 L/hr at 24°C/65% RH to keep a 20% safety margin.
- Setpoint: 22°C air temperature, 65% RH (below 70% prevents powdery mildew; above 60% keeps stomata open).

4. Spatial plan — Warehouse bay (100 m²)

top_down, plan_type=floor, authored by fang, generated 25 Apr 2026, 06:11 (container_linear v1.0.0)

X — envelope length, mm (0 '10000)

Y — envelope depth, mm (0 '10000)



■ Floor-mounted ■ Wall-mounted ■ Ceiling-mounted (above) □ Envelope

Placements					Constraints
Module	Mount	W×D×H mm	x,y,z mm	Rotation °	No constraints recorded.
Port-Side Hydroponic Grow Racks	floor	6860×1200×2700	50,1675,0	0	
Star-board-Side Hydroponic Grow Racks	floor	6860×1200×2700	6960,1675,0	0	
Nutrient Reservoir & Recirculation Skid	floor	2000×1200×2000	13870,1675,0	0	
HVAC & CO ₂ Enrichment Unit	floor	1500×1000×1800	15920,1775,0	0	
LED Grow Light System	ceiling	6860×1200×120	1570,4400,5880	0	
IoT Sensor Mesh & Farm Controller	wall	800×400×1800	7350,0,1000	0	
Insulated Container Shell	envelope	10000×10000×6000	0,0,0	0	
Electrical Distribution Panel	envelope	10000×10000×6000	0,0,0	0	

Notes

- Port-side row overflows envelope by 7470mm — consider symmetric layout or larger envelope.
- Symmetric layout: no (single-row floor).
- Aisle: 1000mm centred along container depth (10000mm interior).
- Target — CANOPY_M2: 200 · TIERS: 4.

3.1 · Insulated Container Shell

Module id: container_shell

4800.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Provides the ISO 668 40ft HC structural envelope, thermal insulation, and hygienic food-grade interior lining that houses all internal subsystems.

WHY IT MATTERS

The shell defines every internal dimension, dictates thermal and humidity stability for the entire grow environment, and provides the only structural anchorage for the racks (each carrying ~750 kg watered) and utility skid. A leak, thermal bridge, or anchor pull-out compromises the whole system; the shell is also the sole barrier against UK ambient conditions ranging from -10 to +35 °C and is the regulatory boundary for food-contact hygiene.

DESCRIPTION

The Insulated Container Shell is a fit-out of a one-trip 40ft High Cube ISO 668 container forming the structural, thermal, and hygienic envelope of the vertical farm. The Corten weathering-steel corrugated shell provides the primary load path (corner castings, bottom rails, cross-members), allowing the unit to be lifted, road-transported on a flatbed, and stacked or sited on level pads without auxiliary structure. Internally, 50 mm rigid PIR sandwich panels are mechanically fixed to walls and ceiling using thermal-break Z-profiles to minimise cold-bridging through the corrugations, achieving an effective U-value of approximately 0.45 W/m²K and reducing parasitic HVAC load to meet the <250 kWh/day energy budget.

KEY PARTS (7)

- One-trip 40ft High Cube ISO 668 Corten steel container shell, 12,192 × 2,438 × 2,896 mm external, ~3,800 kg tare, ISO 1496-1 compliant corner castings
- PIR sandwich insulation panels, 50 mm thick, $U \leq 0.020 \text{ W/m}^2\text{K}$, food-grade FM-approved facers, mechanically fixed to interior walls and ceiling with thermal-break Z-profiles
- 304 stainless steel interior lining sheet, 0.8 mm, 2B finish, sealed with food-grade silicone at all seams (IP65 wash-down rated)
- Reinforced floor system: 28 mm marine ply over existing container floor, with 6 mm hot-rolled steel load-spreader plates under rack and reservoir footprints, anodised aluminium coved skirting
- Insulated personnel door, 900 × 2,000 mm, foamed-core, with magnetic gasket seal and lever latch, replacing one rear container door leaf
- M12 stainless rack anchor inserts (×64), embedded through floor reinforcement to container cross-members, 5 kN pull-out rated each
- GRP drip tray and floor membrane, fully welded seams, 50 mm upstand, drained to utility room sump

FAILURE MODES (4)

- Condensation and mould growth at thermal bridges where insulation meets corrugations or door frame, contaminating the food-grade environment
- Floor deflection or anchor pull-out under fully-watered rack loads (~6 tonnes distributed) leading to misalignment of nutrient drainage gradients
- Gasket and silicone seam degradation under repeated wash-down cycles causing water ingress into insulation core and loss of R-value
- Corten shell corrosion at floor-to-wall junction where standing nutrient solution contacts unprotected steel

OPEN QUESTIONS (3)

- Confirmed fully-laden operational weight including reservoirs (source 5,000 kg figure appears low; likely 8–10 tonnes) and resulting floor reinforcement spec
- Whether a separate insulated personnel door or modified rear cargo doors better serves weekly planting workflow
- Required fire rating of insulation core (PIR vs mineral wool) under UK Building Regs Approved Doc B for an occupied food-production container

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
40ft High Cube ISO container shell — One-trip container UK market price	make	Manual/Assembly · Other	£2,800
PIR sandwich insulation panels, 50mm (×12 panels) — Insulation UK supplier for container refit	make	Manual/Assembly · Other	£480
Thermal-break Z-profiles (×30) — Standard Z-profile for insulation fixing	buy	—	£90
304 stainless steel lining sheet, 0.8mm, 2B finish — Sheet metal from UK stainless supplier	buy	—	£650
Food-grade silicone sealant (10 tubes) — RS Components food-grade silicone	buy	—	£50
28mm marine plywood floor (2 sheets) — Marine ply from builders merchant	buy	—	£160
6mm hot-rolled steel load-spreader plates (×4) — Laser cut and folded, low complexity	make	Sheet Metal · Mild Steel	£120
Anodised aluminium coved skirting (×4 lengths) — Extruded skirting available off-shelf	make	Manual/Assembly · Aluminum	£80
Insulated personnel door, 900x2000mm — Container door kit from specialist	buy	—	£500
M12 stainless rack anchor inserts (×64) — RS Components M12 threaded inserts	buy	—	£320
GRP drip tray and floor membrane — Custom GRP fabrication with welded seams	make	Lay-up GRP · Fibreglass	£350
Labour	£1,600 — 2 fitters × 5 days at £40/hr for installation and sealing		
Per-unit total	£7,200		

ASSUMPTIONS

- Container purchased one-trip
- Insulation fitted by professional team
- GRP tray custom made to size

Container shell with insulation and lining is standard conversion, parts mainly buy except steel plates and GRP tray

3.2 · Port-Side Hydroponic Grow Racks

Module id: grow_racks_port

420.00 kg · 8 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Multi-tier vertical hydroponic racking on the port side of the central aisle holding half of the 6,000 plant slots in NFT/DWC trays.

WHY IT MATTERS

These racks physically define the productive volume of the farm — they hold half of the 6,000 plant slots and therefore directly govern yield, plant spacing, canopy uniformity, and the precision of nutrient and light delivery. Their geometry determines aisle clearance, LED-to-canopy distance (and thus DLI efficiency), and serviceability during weekly plantings. Any structural, hygienic, or drainage failure here propagates immediately to crop loss and downtime.

DESCRIPTION

The port-side grow rack assembly is a freestanding multi-tier hydroponic structure occupying roughly 10 m of the container's longitudinal cultivation zone on the port side of the central aisle. It comprises 5 rack bays, each with 8 growing tiers at 600 mm vertical pitch, providing 3,000 plant sites across 40 NFT-style trays (1,700 x 600 mm). The primary frame uses bolted aluminium T-slot extrusion for corrosion resistance, hygienic wash-down, and field reconfigurability, with stainless steel hardware throughout. Each tier consists of a stainless drip pan supporting a food-safe HDPE NFT channel; nutrient solution enters at the high end via a 16 mm LDPE feed line from the port distribution header, flows as a thin film past plant roots, and exits through a 25 mm spigot into the common return manifold routed along the floor back to the nutrient skid.

KEY PARTS (7)

- Vertical upright frame: 40x40 mm anodised aluminium extrusion (e.g. Bosch Rexroth or Item profile), 2,500 mm tall, 4 uprights per rack bay, floor-anchored via M12 stainless steel bolts to reinforced floor pads
- NFT grow trays: food-safe HDPE channels, 1,700 mm L x 600 mm W x 60 mm deep, 1.5% drainage slope, removable lid with 36 plant site holes per tray (net pot Ø50 mm at 150 mm pitch)
- Tier shelves: 1.5 mm 304 stainless steel drip pans, 1,720 x 620 mm, with rolled edge and integrated 25 mm drainage spigot to common return manifold
- Tier supports: powder-coated steel cross-beams, 600 mm pitch vertically (8 tiers), bolted to upright slots for adjustable canopy height
- Return drainage manifold: 50 mm food-grade PVC-U pipe running floor-level along port wall with 25 mm tee branches per tier, 1% gradient toward nutrient skid return
- Tray retention clips: 316 stainless steel spring clips securing trays against accidental dislodgement during weekly planting
- Cable/tubing tray: aluminium C-channel mounted to upright rear face, routes LED power and nutrient feed lines clear of canopy

FAILURE MODES (4)

- Tray drainage blockage from root mass or biofilm causing flooding, anaerobic conditions, and crop loss
- Frame deflection or racking under wet-tray load (>15 kg/tray fully planted) leading to misaligned LED-to-canopy distances and uneven growth
- Galvanic corrosion at aluminium/stainless interfaces in the high-humidity environment, compromising structural integrity over time
- Biofilm and algal growth on tray surfaces creating food-safety non-conformance and pathogen reservoirs

OPEN QUESTIONS (3)

- Final choice between NFT and shallow DWC tray architecture — affects tray depth, water volume per tier, and reservoir sizing
- Required floor reinforcement detail beneath rack uprights given container baseline floor rating of ~0.5 kN/m²
- Aisle clearance specification (600 vs 800 mm) and its impact on rack depth and tray length tolerance

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
40x40mm anodised aluminium extrusion (×4 uprights) — Item profile from make RS Components		Extrusion · Aluminum	£200
M12 stainless bolts for floor anchor (×16) — RS Components M12 bolts	buy	—	£40
HDPE NFT grow trays (×8) — Standard NFT tray from hydroponics supplier	make	Manual/Assembly · Polyethylene	£320
304 stainless steel drip pans (×8) — Custom size with rolled edge and spigot	make	Sheet Metal · Stainless 304	£480
Powder-coated steel cross-beams (×16) — Cut and welded from steel section	make	Welding · Mild Steel	£320
50mm food-grade PVC-U return drain pipe — Pipe and fittings from plumbing supplier	make	Manual/Assembly · PVC	£60
25mm tee branches (×8) — PVC-U fittings from RS	buy	—	£24
316 stainless steel spring clips (×64) — Standard spring clips from industrial supplier	buy	—	£32
Aluminium C-channel cable tray (×4 lengths) — Standard profile from metal stockist	make	Manual/Assembly · Aluminum	£40
Labour	£800 — 1 fitter × 5 days at £40/hr for rack assembly		
Per-unit total	£2,316		

ASSUMPTIONS

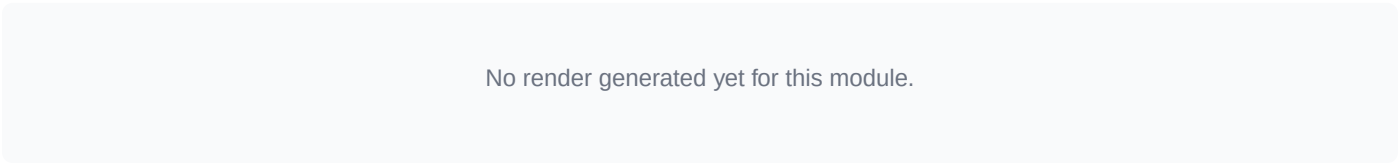
- Extrusion is standard item profile
- Steel parts fabricated locally
- Trays are COTS hydroponic items

Rack structure combines aluminium extrusions with fabricated steel and stainless parts

3.3 · Starboard-Side Hydroponic Grow Racks

Module id: grow_racks_starboard

420.00 kg · 8 wk lead · Specialist judgement · mirrors grow_racks_port



PURPOSE

Multi-tier vertical hydroponic racking on the starboard side of the central aisle holding the other half of the 6,000 plant slots.

WHY IT MATTERS

This module physically holds half the crop and defines plant spacing, light coupling distance, root-zone hydraulics, and airflow paths. Any geometric or hydraulic asymmetry versus the port racks would create yield variation across the container, undermining the 4 t/year target and the uniform DLI/CO₂ control strategy.

DESCRIPTION

The starboard grow rack assembly is a free-standing, floor-anchored multi-tier hydroponic structure occupying roughly 10 m of the container's longitudinal cultivation zone on the starboard side of the central aisle. It comprises 4 racks, each with 6 growing tiers at 380 mm vertical pitch, supporting NFT-style HDPE grow trays sized to a 1,700 x 600 mm module to hold 3,000 plant slots in total. Uprights are anodised aluminium extrusion (corrosion-resistant in the high-RH environment, slotted for tier-height adjustment), with powder-coated steel tier pans lined with food-grade 304 stainless splash liners to handle nutrient overspray and wash-down.

KEY PARTS (6)

- Anodised aluminium extrusion uprights, 40x40 mm slotted profile, 2,400 mm tall, 6 columns per rack
- NFT grow trays, food-grade HDPE, 1,700 x 600 x 60 mm, twin-channel with 2% drainage slope, 25 trays per rack tier
- Tier shelves, powder-coated 1.5 mm steel pan with stainless 304 splash liner, 1,720 x 620 mm, 6 tiers per rack at 380 mm pitch
- Floor-anchor base plates, 200 x 200 x 8 mm galvanised steel with M12 chemical anchors into reinforced floor pads
- Tray-level nutrient distribution manifold, 25 mm food-grade PE pipe with 6 mm dripper take-offs per tray
- Return gutter, sloped uPVC channel 75 mm dia. running base of each rack to common drain header

FAILURE MODES (4)

- Tray slope drift causing flooding or dry spots, leading to root death or algal contamination
- Galvanic or chloride corrosion at steel/aluminium/stainless interfaces in persistent high-RH/nutrient-mist environment
- Floor anchor pull-out or rack sway under transport vibration during flatbed delivery
- Biofilm build-up in return gutters causing cross-tier disease transfer

OPEN QUESTIONS (3)

- Final tier pitch trade-off between LED-to-canopy distance and total tier count to hit exactly 3,000 slots
- Whether NFT or shallow DWC is chosen as the hydroponic technique (affects tray depth and return flow rate)
- Aisle clearance dimension once port and starboard racks are both installed (target 700 mm minimum)

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
40x40mm anodised aluminium extrusion uprights (x6) — Item profile from make RS Components		Extrusion · Aluminum	£300

HDPE NFT grow trays (×25) — Standard hydroponic tray from supplier	make	Manual/Assembly · Polyethylene	£750
Powder-coated 1.5mm steel pan shelves (×6) — Custom size with formed edges	make	Sheet Metal · Mild Steel	£420
304 stainless splash liners (×6) — Laser cut and folded stainless sheets	make	Sheet Metal · Stainless 304	£150
200x200x8mm galvanised steel base plates (×6) — Cut from plate and drilled	make	Plating · Zinc	£60
M12 chemical anchors (×12) — RS Components chemical anchor kit	buy	—	£60
25mm food-grade PE pipe and dripper take-offs — Hydroponic irrigation fittings	make	Manual/Assembly · Other	£80
75mm uPVC return gutter — Standard uPVC pipe and fittings	buy	—	£40
Labour	£600 — 1 fitter × 3.75 days at £40/hr for assembly and plumbing		
Per-unit total	£2,460		

ASSUMPTIONS

- Extrusion profiles standard
- Steel parts fabricated locally
- Chemical anchors for floor fixing

Rack similar to port side with fewer tiers but more trays, uses fabricated steel and stainless parts

3.4 · LED Grow Light System

Module id: led_lighting_system

230.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Full-spectrum tier-mounted LED arrays delivering DLI-controlled photosynthetic light to every growing level on both sides.

WHY IT MATTERS

Lighting is the single largest energy consumer in the container (typically 60–70 % of the <250 kWh/day budget) and is the primary determinant of yield, crop quality, and morphology. Achieving the 4 t/year target within the energy cap requires high photon efficacy (>2.6 $\mu\text{mol/J}$) and accurate DLI control; any shortfall directly reduces saleable biomass or breaches the energy budget.

DESCRIPTION

The LED grow light system is a distributed array of full-spectrum horticultural LED bars mounted to the underside of every growing tier above the canopy of the tier below, sized to the 1,700 × 600 mm tray footprint. Each bar uses high-efficacy (~2.7 $\mu\text{mol/J}$) mid-power white LEDs supplemented with 660 nm deep-red emitters in an IP65 anodised aluminium extrusion that doubles as a passive heatsink, allowing fanless operation in the high-humidity grow chamber. Bars are powered by remote constant-current drivers (Mean Well HLG-class or equivalent) located in a ventilated driver tray on the rack uprights so that heat from driver losses can be extracted by the HVAC return path rather than dumped at the canopy.

KEY PARTS (5)

- Full-spectrum LED grow bars, 1,680 mm length × 60 mm width, 120 W per bar, ~2.7 $\mu\text{mol/J}$ efficacy, white + deep red (660 nm) spectrum, IP65 rated aluminium extrusion housing — quantity 64–96 (one or two bars per tier per rack across ~48 tier-positions)
- Constant-current LED drivers, 240 W class, 0–10 V / DALI dimmable, 200–240 V AC input, IP67 (e.g. Mean Well HLG-240H series) — quantity matched to bar count, mounted in driver tray on rack uprights
- Dimming and control wiring loom: 0–10 V or DALI bus daisy-chained per zone, plus 3-core H07RN-F flexible mains drops with IP67 quick-connect couplers (Wieland GST18 or equivalent) for tool-free service
- Suspension and reflector hardware: anodised aluminium mounting brackets, stainless M6 fixings, polycarbonate side reflectors to direct photons onto canopy and reduce aisle spill
- Per-rack zone controller / DALI gateway providing photoperiod scheduling, dimming, and fault reporting back to the farm controller

FAILURE MODES (4)

- Driver failure from humidity ingress or thermal cycling, taking out an entire tier zone until replaced
- Gradual photon flux degradation (LM-80/L90 decay) reducing DLI below crop setpoint and silently cutting yield if not compensated by the controller
- Corrosion of connectors and mains drops in the high-RH, nutrient-mist environment leading to earth faults and RCD trips
- Hotspot formation or non-uniform PPFD distribution at canopy edges causing uneven crop growth and harvest losses

OPEN QUESTIONS (3)

- Final bar count, wattage, and spectrum recipe per crop (lettuce vs herbs vs brassicas) needed to hit target DLI within the lighting energy sub-budget

- Whether 0–10 V analogue dimming is sufficient or full DALI-2 addressable control is required for per-tier tuning and fault reporting
- IP rating margin and conformal coating strategy required for long-term reliability at 70–80 % RH with periodic wash-down

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Full-spectrum LED grow bars, 1,680 mm × 60 mm, 120 W, IP65 — COTS grow bars, bulk price £200-300 each, 64-96 bars	buy	—	£19,200
Constant-current LED drivers, Mean Well HLG-240H, dimmable, 240 W, IP67 — COTS Mean Well drivers, ~£70 each, 64-96 units	buy	—	£4,480
Dimming and control wiring loom (0-10 V / DALI, H07RN-F, Wieland couplers) — COTS cables, connectors per zone, ~£800 total	buy	—	£800
Anodised aluminium mounting brackets — Custom brackets, laser cut + fold, £5-10 each × 64-96	make	Sheet Metal · Aluminum	£480
Stainless M6 fixings — COTS fasteners, £0.1-0.2 each, ~500 needed	buy	—	£96
Polycarbonate side reflectors — Custom extruded/cut sheet, £3-4 each × 64-96	make	Laser Cutting · Polycarbonate	£288
Per-rack zone controller / DALI gateway — COTS DALI controller, ~£200 each × 3-4 racks	buy	—	£600
Labour	£960 — Install bars, drivers, wiring, brackets: 24 hours × £40/hr		
Per-unit total	£26,904		

ASSUMPTIONS

- 64 bars, 64 drivers, 3 controllers
- Bulk pricing discount applied

Combined COTS and custom parts for 64-96 bar LED system

3.5 · Nutrient Reservoir & Recirculation Skid

Module id: nutrient_system

180.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Stores, filters, doses (pH/EC), and recirculates hydroponic solution to all rack tiers via a closed-loop pump and pipework system.

WHY IT MATTERS

This module is the single point of failure for crop nutrition and water hygiene: it determines yield, water-use efficiency (recirculation enables >90% water reuse vs. open hydroponics), and biosecurity. A dosing or pH excursion can wipe out a full 6,000-slot crop cycle within 24 hours, directly impacting the 4 t/year yield target and CapEx payback.

DESCRIPTION

The nutrient skid is a pre-assembled, palletised wet-system module that occupies roughly 1.4 m² of the ~4 m² utility room at the door end of the container. It comprises an 800 L opaque HDPE main reservoir, a vertical multistage recirculation pump delivering ~5 m³/h at 4 bar through a 32 mm PVC-U distribution manifold to port and starboard rack feeds, and a return loop with strainer, cartridge filter, UVC steriliser, and inline pH/EC flow cell. Four diaphragm metering pumps draw from 25 L concentrate tanks (pH-up, pH-down, Nutrient A, Nutrient B) and inject downstream of the main pump under closed-loop control from the farm controller, targeting EC 1.2–2.4 mS/cm and pH 5.5–6.5 depending on crop recipe.

KEY PARTS (9)

- Main nutrient reservoir: 800 L food-grade HDPE tank, opaque black to inhibit algal growth, with bunded drip tray and lockable lid (~1200×800×900 mm, fits within 4 m² utility room footprint)
- Primary recirculation pump: Grundfos CRE 5-4 or equivalent vertical multistage centrifugal, 3-phase, ~1.5 kW, 5 m³/h at 4 bar, EPDM seals, 316SS wetted parts
- Dosing pump skid: 4× Grundfos DDA 7.5-16 digital diaphragm metering pumps (pH-up KOH, pH-down H₃PO₄, Nutrient A, Nutrient B), with 25 L PE concentrate tanks and lance suction assemblies
- Inline pH and EC sensors: Hanna HI1001 industrial pH probe + HI7635 EC probe with 4-20 mA transmitters, mounted in a flow cell on the return manifold
- Particulate filtration: 100 µm stainless mesh return strainer + 50 µm polypropylene cartridge filter housing (BSP 1" connections)
- UV sterilisation unit: 55 W low-pressure amalgam UVC inline steriliser (e.g. Aquafine OptiVenn) for pathogen suppression in recirculating loop
- Distribution manifold: 32 mm food-grade PVC-U manifold with solenoid-actuated zone valves (one per rack tier zone), pressure-compensating drippers/NFT inlets
- Float and ultrasonic level sensors in main reservoir, plus solenoid-actuated mains top-up valve with RPZ backflow preventer to comply with WRAS Fluid Category 5
- Skid frame: powder-coated 40×40 mm mild steel box section, ~1.4×1.0 m, with integrated drip tray and forklift pockets for install

FAILURE MODES (4)

- Root-zone pathogen outbreak (Pythium, Fusarium) due to UV lamp degradation or filter bypass, contaminating the entire recirculating loop
- Dosing pump runaway or stuck check-valve causing pH crash or EC overshoot, killing crops before alarm thresholds trigger
- Pump cavitation or blockage from accumulated biofilm/root debris in return strainer, starving upper tiers of feed

- Reservoir or pipework leak onto container floor, leading to electrical hazard and structural corrosion of the steel container shell

OPEN QUESTIONS (3)

- Whether NFT, DWC, or ebb-and-flood is the chosen rack hydraulic regime — this changes flow rate, head, and reservoir sizing materially
- UK water authority requirements for backflow protection on mains top-up (WRAS Fluid Category 5 likely mandates an AB air gap rather than RPZ)
- Disposal/refresh strategy for spent nutrient solution — full dump cycle frequency drives effluent handling and water consumption

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
800L HDPE tank with bund and lid — COTS food-grade tank, ~£1200	make	Manual/Assembly · Polyethylene	£1,200
Grundfos CRE 5-4 recirculation pump, 1.5 kW — COTS pump, ~£1800	buy	—	£1,800
4× Grundfos DDA 7.5-16 dosing pumps with 25L tanks and lances — COTS dosing pumps, ~£1500 each	make	Manual/Assembly · Other	£6,000
Hanna HI1001 pH probe — COTS sensor, ~£350	buy	—	£350
Hanna HI7635 EC probe — COTS sensor, ~£250	buy	—	£250
4-20 mA transmitters for sensors — COTS transmitters, ~£100 each × 2	buy	—	£200
Flow cell for sensors — COTS flow cell, ~£100	buy	—	£100
100 µm stainless mesh return strainer — COTS strainer, ~£80	buy	—	£80
50 µm polypropylene cartridge filter housing — COTS housing with cartridge, ~£120	make	Manual/Assembly · Other	£120
Aquafine OptiVenn 55 W UV steriliser — COTS UV unit, ~£1200	buy	—	£1,200
32 mm PVC-U distribution manifold with solenoid valves — COTS manifold kit, ~£400	buy	—	£400
Pressure-compensating drippers/NFT inlets — COTS drippers, ~£200	buy	—	£200
Float and ultrasonic level sensors — COTS sensors, ~£150 total	buy	—	£150
Solenoid top-up valve with RPZ — COTS WRAS-compliant valve, ~£300	buy	—	£300
Powder-coated mild steel skid frame, 1.4×1.0 m — Custom welded frame with drip tray, ~£400	make	Welding · Mild Steel	£400
Labour	£800 — Assemble skid, pipework, wiring, tanks: 20 hours × £40/hr		
Per-unit total	£13,350		

ASSUMPTIONS

- Single skid per farm unit
- Bulk pricing for pumps not included

Mostly COTS pumps, sensors, piping; custom frame

3.6 · HVAC & CO, Enrichment Unit

Module id: hvac_co2

220.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Maintains grow-room temperature, humidity, air filtration, and CO₂ enrichment setpoint by conditioning recirculated air.

WHY IT MATTERS

Climate control is the single largest determinant of crop quality, disease pressure, and yield consistency: leafy greens tip-burn, bolt, or develop powdery mildew within hours if VPD drifts. CO₂ enrichment to ~1,000 ppm typically lifts photosynthesis and yield by 20-30%, directly impacting the 4 t/year target. The HVAC also represents the second-largest energy load after lighting and must stay within the <250 kWh/day envelope, making COP and latent-handling efficiency critical to the business case.

DESCRIPTION

The HVAC & CO₂ enrichment unit is a packaged direct-expansion air-conditioning system specifically tuned for high-latent-load horticultural environments, combined with a regulated CO₂ injection skid. The indoor air handler is wall-mounted at the utility-room end of the container and draws return air from a low-level plenum, passes it through pre/fine filtration and a UV-C germicidal section, then through DX cooling and hot-gas reheat coils to independently control sensible temperature (target 20-22 °C) and dehumidification to a 65 ±5% RH setpoint despite ~80-120 L/day of plant transpiration. An EC plug fan delivers ~2,000 m³/h through a 250 mm insulated spiral duct running along the container roof apex, distributing supply air via fabric sock diffusers to ensure uniform velocity (0.3-0.5 m/s) across all canopy levels on both port and starboard racks.

KEY PARTS (7)

- Ducted split-system DX air handler, ~12 kW sensible + 8 kW latent cooling, R454B refrigerant, EC plug fan ~2,000 m³/h, stainless casing for wash-down
- External condensing unit, 14 kW nominal, mounted on container roof or end-wall plinth, low-ambient kit for UK winter operation down to -10 °C
- Hot-gas reheat / dehumidification coil with electronic expansion valve to handle 60-75% RH setpoint under high transpiration load
- G4 pre-filter + F7 fine filter + UV-C in-duct lamp (Philips TUV 36W or equivalent) for airborne pathogen control
- CO₂ enrichment subsystem: 2 × 34 kg liquid CO₂ cylinders, two-stage regulator, solenoid shut-off, 0-10 V mass-flow controller (Aalborg GFC17), stainless 6 mm distribution tubing to in-duct sparger
- NDIR CO₂ sensor (Vaisala GMP252, 0-2,000 ppm), SHT35 T/RH sensors (×4 distributed), all on Modbus RTU bus to farm controller
- Insulated galvanised spiral supply duct 250 mm dia. with perforated textile sock diffusers along aisle, return plenum at opposite end of container

FAILURE MODES (4)

- Condensate drain blockage causing water ingress into electrical panel and biofilm/Legionella growth in pan
- Refrigerant leak from corroded coil in high-humidity wash-down environment, leading to loss of cooling and crop loss within 12 hours
- CO₂ regulator or solenoid failure in the open position, creating an asphyxiation hazard for service personnel entering the container
- Filter loading not detected, causing airflow drop, hotspots in canopy, and elevated humidity triggering fungal disease outbreak

OPEN QUESTIONS (3)

- Peak simultaneous sensible + latent load envelope under full lighting at lights-on transition — needs CFD or psychrometric modelling against confirmed transpiration rate
- Whether a single split system meets redundancy expectations for unattended weekly operation or if N+1 modular units are required within budget
- UK F-gas regulation impact on refrigerant choice (R454B vs R32) and certified installer requirements for serial production

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
DX air handler, 12 kW sensible + 8 kW latent, EC fan 2000 m³/h, stainless buy — COTS air handler, custom stainless option, ~£4500	—		£4,500
Condensing unit 14 kW, low-ambient kit — COTS condensing unit, ~£3500	buy	—	£3,500
Hot-gas reheat coil with EEV — COTS reheat package, ~£800	buy	—	£800
G4 and F7 filter set — COTS filters, ~£200	buy	—	£200
UV-C lamp, Philips TUV 36W with holder — COTS UV lamp, ~£80	buy	—	£80
2 × 34 kg CO, cylinders with two-stage regulator— COTS cylinders and regulator, ~£600	buy	—	£600
Solenoid shut-off valve for CO,— COTS valve, ~£100	buy	—	£100
Aalborg GFC17 0-10 V mass-flow controller — COTS MFC, ~£600	buy	—	£600
Stainless 6 mm CO, distribution tubing, 50 m— COTS tubing, ~£150	buy	—	£150
Vaisala GMP252 NDIR CO, sensor— COTS CO, sensor, ~£400	buy	—	£400
SHT35 T/RH sensors ×4 — COTS sensors, ~£30 each	buy	—	£120
Modbus RTU interface — COTS Modbus module, ~£50	buy	—	£50
250 mm insulated galvanised spiral duct, 10 m — COTS duct and insulation,make ~£300		Plating · Zinc	£300
Perforated textile sock diffusers, 10 m — COTS diffuser socks, ~£250	buy	—	£250
Labour	£1,200 — Install ducting, refrigerant lines, sensors: 30 hours × £40/hr		
Per-unit total	£12,850		

ASSUMPTIONS

- Single unit per container
- Refrigerant line sets separate

All COTS HVAC, CO, enrichment, and sensor components

3.7 · Electrical Distribution Panel

Module id: electrical_distribution

55.00 kg · 8 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Receives the three-phase mains supply and distributes protected, switched power feeds to every powered subsystem within the container.

WHY IT MATTERS

It is the single point of electrical safety and energy control for the entire farm: every other subsystem depends on it for protected, switched power, and any fault here takes the whole crop offline. It also enforces the <250 kWh/day energy budget through sub-metering and contactor-based load shedding commanded by the IoT controller.

DESCRIPTION

A wall-mounted three-phase distribution and control panel installed in the ~4 m² utility room at the door end of the container. It receives a 400V 50Hz three-phase TN-S mains supply (sized for ~60-80A per phase to support the <250 kWh/day load with headroom) via a 5-core armoured tail and CEE 63A or hardwired gland entry. Behind the lockable IP54 enclosure, an incoming isolator, RCD and Type 2 SPD feed a busbar that distributes to MCB-protected sub-circuits: LED driver banks (largest single load, ~80-100 kWh/day), the HVAC compressor and fans, nutrient and dosing pumps, the CO, solenoid, and a 24VDC PSU rail for the controller and sensor mesh.

KEY PARTS (8)

- Main wall-mounted IP54 metal enclosure, ~800x600x250mm, powder-coated steel, with hinged lockable door (e.g. Rittal AE or Schneider Spacial S3D)
- 125A 4-pole main isolator switch with lockable handle (e.g. ABB OT125F4) rated for 400V 50Hz three-phase mains
- Type B 4-pole 125A surge protection device (SPD) and 300mA Type A RCD on incomer for shock and transient protection
- Bank of 3-phase and single-phase MCBs (Schneider iC60N or equivalent): dedicated circuits for LED drivers (~120A across 3 phases), HVAC compressor (32A 3ph), pumps/dosing (16A 1ph), controller/sensor 24V PSU (6A 1ph), socket outlets and lighting
- DIN-rail 24VDC 240W switched-mode power supply (e.g. Meanwell DRP-240-24) for controller and sensor mesh
- Three-phase energy meter with Modbus RTU output (e.g. Schneider iEM3155) for sub-circuit energy monitoring
- Contactor bank (4-pole 40A, e.g. ABB AF series) for controller-commanded switching of LED and HVAC circuits
- Phoenix Contact terminal blocks, ferruled tinned-copper wiring (H07V-K), and 35mm² copper earth bar bonded to container shell

FAILURE MODES (4)

- Nuisance RCD trip from cumulative LED driver leakage current causing total crop loss within hours
- Contactor weld-on under inrush from large LED driver banks, preventing scheduled lights-off and overheating the grow zone
- Condensation ingress in humid grow environment causing corrosion of busbars and terminal blocks
- Phase imbalance from poorly distributed single-phase loads tripping the incomer or overheating one phase conductor

OPEN QUESTIONS (3)

- Final per-phase load schedule and whether 63A or 100A incomer is required once LED driver efficiency is fixed
- Whether site supply is TN-S, TN-C-S or TT — affects earthing arrangement and RCD selection
-

Need for an integrated UPS or generator changeover to ride through short outages and protect the nutrient pumps and controller

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Main enclosure (Rittal AE 800x600x250mm) — Standard IP54 steel enclosure, RS price.	make	Manual/Assembly · Other	£450
125A 4-pole main isolator (ABB OT125F4) — COTS switch, Farnell price.	buy	—	£120
Type B 4-pole 125A SPD — Standard surge protector, RS price.	buy	—	£80
300mA Type A RCD 4-pole — COTS residual current device.	buy	—	£60
MCBs (Schneider iC60N) various ratings — 8 MCBs at ~15 each, RS price.	buy	—	£120
24VDC 240W PSU (Meanwell DRP-240-24) — DIN-rail supply, RS price.	buy	—	£85
Three-phase energy meter (Schneider iEM3155) — Modbus metering, RS price.	buy	—	£110
Contactor bank 4-pole 40A (ABB AF) — Two contactors at ~45 each.	buy	—	£90
Terminal blocks, wiring, earth bar — Phoenix connectors, cable, Farnell.	buy	—	£60
DIN rail, labels, cable glands — Standard assembly accessories.	buy	—	£30
Labour	£200 — 5 hours wiring/assembly at £40/hr.		
Per-unit total	£1,405		

ASSUMPTIONS

- All parts COTS standard.
- No custom sheet metal work needed.

Electrical distribution panel assembled from standard industrial components.

3.8 · IoT Sensor Mesh & Farm Controller

Module id: control_iot

35.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Distributed sensor network and central controller/gateway that orchestrates lighting, climate, nutrient dosing, and uplinks data to the remote dashboard.

WHY IT MATTERS

Unattended weekly operation is only viable if the controller reliably orchestrates lighting, climate, and dosing without drift, while flagging anomalies to the remote operator before crop loss occurs. A single missed pH excursion or lighting failure can write off a 6,000-slot batch worth thousands of pounds, so this module directly underwrites yield consistency, energy budget compliance (<250 kWh/day), and the commercial viability of remote fleet operation.

DESCRIPTION

The IoT Sensor Mesh & Farm Controller is the digital nervous system of the container farm. It comprises a central industrial edge-controller (PLC or Linux-based controller such as Revolution Pi) housed in an IP65 stainless or powder-coated steel cabinet in the utility room, a 7-inch HMI for local operation, and a wireless mesh of low-power sensor nodes distributed across both rack arrays and the nutrient skid. The controller runs deterministic control loops for climate setpoints, DLI integration for lighting, and pH/EC dosing PID loops, while a parallel cloud agent uplinks telemetry over 4G/Ethernet via TLS-secured MQTT to the remote dashboard. Sensor nodes use LoRaWAN or Zigbee mesh radios at 868 MHz / 2.4 GHz, battery- or 24 VDC-powered, with industrial probes (Atlas Scientific or Hanna for pH/EC, Sensirion SCD41 for CO₂, SHT45 for T/RH) feeding either directly via 4–20 mA / RS-485 to the PLC IO or wirelessly to a gateway. Materials are food-safe IP-rated polycarbonate and stainless-steel sensor housings to survive 70% RH wash-down environments.

KEY PARTS (6)

- Industrial PLC / edge controller (e.g. Siemens S7-1200 or Revolution Pi RevPi Connect 4) with DIN-rail mount, Modbus TCP/RTU, MQTT, and 4G/Ethernet uplink
- Distributed wireless sensor nodes (LoRaWAN or Zigbee 2.4 GHz mesh) covering air T/RH at 8–12 grow-zone points, IR leaf-temperature, and tier-level CO₂
- In-line nutrient sensor manifold: industrial pH probe (± 0.05 pH), EC probe ($\pm 2\%$ FS), dissolved O₂, ultrasonic reservoir level, and PT100 solution temperature, all 4–20 mA into IO module
- 24 VDC redundant PSU (Mean Well DRT-240) with UPS backup (12 Ah sealed lead-acid) for graceful shutdown and alarm continuity
- Touchscreen HMI (7-inch IP65) mounted in utility room for local override, plus cloud gateway with TLS-secured MQTT to remote farm-management dashboard
- Sealed IP65 control cabinet (600 × 800 × 250 mm) with 24 V relay/SSR bank for command outputs to LED drivers, dosing pumps, and HVAC contactors

FAILURE MODES (4)

- pH/EC probe drift or fouling causing incorrect dosing and crop damage if not auto-calibrated or flagged
- Loss of 4G/Ethernet uplink leaving the unit operating blind to the remote dashboard; mitigated by local autonomous control but loses visibility
- Wireless mesh interference or node battery exhaustion causing stale telemetry and missed alarms
- Controller firmware lockup or SD-card corruption halting all setpoint commands; requires watchdog and redundant boot media

OPEN QUESTIONS (3)

- Choice between fully wired Modbus/RS-485 sensor bus versus wireless LoRaWAN/Zigbee mesh — trade-off of installation cost vs reliability in a metal container RF environment
- Cybersecurity and OTA update strategy for fleet of remotely-managed units (certificate management, VPN vs MQTT broker)
- Required sensor redundancy level for pH/EC — single probe with calibration schedule or dual-probe voting for fail-safe dosing

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Industrial edge controller (Siemens S7-1200) — COTS PLC with Ethernet, RS price.	buy	—	£500
LoRaWAN sensor nodes (12 units) — ~50 each, Farnell, for T/RH/CO2.	buy	—	£600
pH probe 4-20mA — Industrial inline sensor, RS price.	buy	—	£150
EC probe 4-20mA — Standard conductivity sensor.	buy	—	£130
Dissolved O2 probe 4-20mA — Typical cost, RS/Farnell.	buy	—	£200
Ultrasonic level sensor 4-20mA — Standard reservoir level sensor.	buy	—	£80
PT100 temperature probe 4-20mA — COTS RTD sensor.	buy	—	£40
24VDC redundant PSU (Mean Well DRT-240) — DIN-rail redundant supply.	buy	—	£120
UPS backup 12Ah SLA battery — Sealed lead-acid, RS price.	buy	—	£50
Touchscreen HMI 7-inch IP65 — COTS panel, RS price.	buy	—	£350
IP65 control cabinet 600x800x250mm — Standard enclosure, RS.	buy	—	£400
Relay/SSR bank for outputs — 8 relays, DIN-mount, Farnell.	buy	—	£80
Cables, connectors, DIN rail — Standard wiring accessories.	buy	—	£50
Labour	£320 — 8 hours wiring/assembly/configuration at £40/hr.		
Per-unit total	£3,070		

ASSUMPTIONS

- All sensors COTS.
- No custom metal or plastic parts.

IoT sensor mesh built entirely from off-the-shelf industrial components.

4. BOM master (70 rows)

BOM derived from the module decomposition's keyParts, expanded into typed part rows. Part records live in the `parts` table and are joined back to modules via source_module_id.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
CTL-001	IP65 Control Cabinet — IP65-rated sheet metal control cabinet housing PLC, PSU, relays, gateway, and terminal blocks. Wall-mounted within the container with sealed cable glands and lockable hinged door.	IoT Sensor Mesh & Farm Controller	make	sheet_metal · 304 Stainless Steel · ±±1.0mm	18.00kg	£320
CTL-ASY	IoT Sensor Mesh & Farm Controller Assembly — Top-level assembly comprising the central PLC edge controller, distributed wireless mesh sensor nodes, gateway, and associated mounting and wiring within the container. Orchestrates lighting, climate, dosing, and remote data uplink.	IoT Sensor Mesh & Farm Controller	make	other · Mixed (steel enclosure, PCB, plastics) · ±±1mm (assembly)	8.50kg	£2,800
CTL-PUR-001	Industrial PLC Edge Controller — Industrial PLC edge controller with Ethernet, Modbus, and digital/analog I/O for orchestrating all container subsystems. Runs control logic and provides MQTT uplink to the cloud dashboard.	IoT Sensor Mesh & Farm Controller	buy	—	0.90kg	£850
CTL-PUR-002	Wireless Mesh Sensor Node — Battery- or 24VDC-powered wireless mesh sensor node with onboard temperature, humidity, CO ₂ , and light sensors. Communicates over LoRa/Zigbee mesh to the farm controller gateway.	IoT Sensor Mesh & Farm Controller	buy	—	0.18kg	£95
CTL-PUR-003	Dissolved Oxygen Probe — Industrial dissolved oxygen probe for hydroponic nutrient solution monitoring. Galvanic or optical DO sensor with RS485/4-20mA output for integration with farm controller.	IoT Sensor Mesh & Farm Controller	buy	—	0.25kg	£280
CTL-PUR-004	24VDC Redundant PSU with UPS — DIN-rail mounted 24VDC redundant power supply with integrated UPS module to provide uninterrupted power to controllers, sensors, and HMI during mains dropouts.	IoT Sensor Mesh & Farm Controller	buy	—	2.50kg	£380
CTL-PUR-005	7-inch IP65 Touchscreen HMI — 7-inch IP65-rated capacitive touchscreen HMI for local operator interface, supporting Modbus TCP/EtherNet-IP for control of dosing, lighting, and HVAC subsystems.	IoT Sensor Mesh & Farm Controller	buy	—	1.20kg	£450

CTL-PUR-006	Cloud Gateway Module — Industrial IoT cloud gateway with 4G/LTE and Ethernet up-link, providing secure MQTT/HTTPS connectivity to remote dashboard for telemetry and OTA control.	IoT Sensor Mesh & Farm Controller	buy	—	0.50kg	£320
CTL-PUR-007	24V Relay/SSR Module — DIN-rail 8-channel 24V relay/SSR interface module for switching contactors, solenoids, and dosing pumps from PLC digital outputs with optical isolation.	IoT Sensor Mesh & Farm Controller	buy	—	0.30kg	£65
ELE-001	IP54 Distribution Enclosure — Wall-mount IP54 sheet steel distribution enclosure with hinged lockable door, gland plate, and DIN rail / mounting plate for switchgear installation. Sized to house 125A main switch plus 36 modular outgoing ways.	Electrical Distribution Panel	make	sheet_metal · Mild steel · ±±1.0mm	18.00kg	£240
ELE-ASY	Electrical Distribution Panel Assembly — Complete electrical distribution panel assembly comprising IP54 enclosure, main isolator, SPD/RCD protection, MCBs, contactors, and labelled feeder terminals for all container subsystems. Pre-wired and tested to BS 7671.	Electrical Distribution Panel	make	other · Mixed (steel enclosure, copper busbar, polymer DIN components) · ±Assembly ±2mm	48.00kg	£3,200
ELE-PUR-001	125A 4-Pole Main Isolator Switch — 125A four-pole rotary main isolator switch for three-phase plus neutral supply disconnection. Door-mounted with red/yellow handle and padlockable in OFF position.	Electrical Distribution Panel	buy	—	1.20kg	£95
ELE-PUR-002	Type B SPD and RCD Module — Combined Type B surge protection device and 125A 4-pole 30mA Type A RCD module on DIN rail, providing transient overvoltage and earth-leakage protection for the entire container installation.	Electrical Distribution Panel	buy	—	1.60kg	£285
ELE-PUR-003	MCB (Three-Phase / Single-Phase) — Miniature circuit breaker for branch protection of three-phase and single-phase distribution circuits within the main panel. DIN-rail mounted, Type C trip curve.	Electrical Distribution Panel	buy	—	0.18kg	£28
ELE-PUR-004	24VDC DIN-Rail Power Supply — Regulated 24VDC DIN-rail power supply providing low-voltage control power to PLC, sensors, valves, and contactor coils. 240W output with universal AC input.	Electrical Distribution Panel	buy	—	1.20kg	£110

ELE-PUR-005	Three-Phase Modbus Energy Meter — Three-phase DIN-rail energy meter with Modbus RTU output for real-time monitoring of container power consumption. Provides kWh, voltage, current, and power factor data to the farm controller.	Electrical Distribution Panel	buy	—	0.35kg	£95
ELE-PUR-006	4-Pole 40A Contactor — 4-pole 40A AC contactor with 24VDC coil for switching three-phase loads such as HVAC compressors and large LED banks. DIN-rail mounted.	Electrical Distribution Panel	buy	—	0.45kg	£42
ELE-PUR-007	DIN-Rail Terminal Block — Screw-type DIN-rail terminal block for distribution and termination of power and control wiring. 6mm² rated for general circuit interconnection.	Electrical Distribution Panel	buy	—	0.01kg	£1
GRP-001	NFT Grow Tray (Port) — NFT (Nutrient Film Technique) grow tray with channelled base and integrated planting holes for the port-side rack tier; injection moulded for food-contact use.	Port-Side Hydroponic Grow Racks	make	injection_molding · Polypropylene · ±±0.5mm	4.20kg	£38
GRP-002	Stainless Steel Tier Drip Pan (Port) — Sheet metal drip pan installed beneath each NFT tray to capture leaks and condensate, sloped to the return manifold.	Port-Side Hydroponic Grow Racks	make	sheet_metal · 304 Stainless Steel · ±±0.5mm	6.50kg	£58
GRP-003	Powder-Coated Tier Cross-Beam (Port) — Horizontal cross-beam connecting the rack uprights and supporting trays/drip pans; formed steel C-section bolted to T-slot uprights.	Port-Side Hydroponic Grow Racks	make	sheet_metal · Mild Steel · ±±0.5mm	3.80kg	£22
GRP-004	PVC-U Return Drainage Manifold (Port) — PVC-U gravity drainage manifold collecting return flow from each tier's drip pan and routing it back to the nutrient reservoir.	Port-Side Hydroponic Grow Racks	make	other · PVC-U · ±±1.0mm	5.50kg	£75
GRP-005	Aluminium Cable/Tubing C-Channel Tray — Aluminium C-channel cable tray running along the rack length to organise LED power, sensor cabling, and nutrient feed/return tubing.	Port-Side Hydroponic Grow Racks	make	other · 6063 Aluminium · ±±0.5mm	5.20kg	£65
GRP-ASY	Port-Side Hydroponic Grow Rack Assembly — Top-level assembly for the port-side multi-tier hydroponic grow rack, integrating uprights, tiers, trays, drip pans, drainage, and cable management.	Port-Side Hydroponic Grow Racks	make	other · Mixed (Aluminium / Stainless / PVC-U) · ±±2.0mm assembly	420.00kg	£6,800
GRP-PUR-001	40x40 Aluminium T-Slot Extrusion Upright — Vertical structural upright for the grow rack, made from 40x40mm aluminium T-slot profile providing tier mounting and adjustability.	Port-Side Hydroponic Grow Racks	buy	—	3.60kg	£32

GRP-PUR-002	316 Stainless Tray Retention Clip — Stainless spring clip retaining each grow tray on its cross-beam to prevent shifting during loading/harvest operations.	Port-Side Hydroponic Grow Racks	buy	—		0.03kg	£2
GRS-001	NFT Grow Tray (Starboard) — Injection-moulded NFT (Nutrient Film Technique) grow tray with integrated channels and plant slots, sloped to drain returning solution to the gutter. Food-contact safe and UV-stabilised for grow-light exposure.	Starboard-Side Hydroponic Grow Racks	make	injection_molding · Food-grade Polypropylene · ±±0.5mm		4.80kg	£38
GRS-002	Tier Shelf Powder-Coated Pan (Starboard) — Powder-coated sheet metal pan forming the structural shelf at each grow tier, supporting NFT trays and catching incidental drips. Folded edges add stiffness and provide mounting flanges to the rack uprights.	Starboard-Side Hydroponic Grow Racks	make	sheet_metal · Mild Steel · ±±0.5mm		12.50kg	£55
GRS-003	Galvanised Floor-Anchor Base Plate — Hot-dip galvanised steel base plate bolted to the container floor, anchoring the rack uprights and distributing static and dynamic loads. Pre-drilled with hole pattern for M10 floor anchors and rack post fixings.	Starboard-Side Hydroponic Grow Racks	make	sheet_metal · Mild Steel · ±±1.0mm		3.60kg	£19
GRS-004	Tray-Level Nutrient Distribution Manifold — Tray-level nutrient distribution manifold consisting of a PVC header with multiple drip-emitter outlets, delivering metered solution to the inlet end of each NFT tray. Threaded inlet connects to the recirculation skid feed line.	Starboard-Side Hydroponic Grow Racks	make	other · uPVC · ±±0.5mm		0.90kg	£24
GRS-005	uPVC Return Gutter Channel — uPVC return gutter channel running along the low end of each tier, collecting effluent from NFT trays and routing it back to the reservoir return line. UV-stable and food-contact safe.	Starboard-Side Hydroponic Grow Racks	make	other · uPVC · ±±1.0mm		2.40kg	£15
GRS-ASY	Starboard-Side Hydroponic Grow Rack Assembly — Complete starboard-side hydroponic grow rack assembly comprising multi-tier vertical NFT/DWC trays, structural framing, plumbing manifolds, and floor anchorage. Houses approximately half of the container's 6,000 plant slots.	Starboard-Side Hydroponic Grow Racks	make	other · Mixed (Aluminium frame, food-grade plastics, steel anchors) · ±±2mm (assembly)		420.00kg	£8,500
HVC-001	Insulated Spiral Supply Duct with Textile Diffuser — Spiral wound galvanised supply duct with 25mm closed-cell insulation and fabric textile diffuser for low-velocity uniform air distribution along the central aisle. Sized for ~2000 m³/h supply airflow.	HVAC & CO, Enrichment Unit	make	sheet_metal · Galvanised steel with PE foam insulation and polyester textile · ±±2mm diameter, ±5mm length		22.00kg	£480

HVC-ASY	HVAC & CO2 Enrichment Unit Assembly — HVAC & CO, Enrich-make Top-level assembly integrating the ducted DXment Unit air handler, condenser, reheat coil, filtration, UV-C disinfection, and CO2 enrichment sub-systems into a single climate control package for the grow room.	other · Mixed As- sembly · ±±5mm	220.00kg	£8,500
HVC-PUR-001	Ducted DX Air Handler — Ducted direct ex- HVAC & CO, Enrich-buy pansion (DX) air handling unit providing re- ment Unit circulation, cooling, and dehumidification for the grow room. Fan section delivers condi- tioned air to the supply duct network.	—	95.00kg	£2,800
HVC-PUR-002	External Condensing Unit — External HVAC & CO, Enrich-buy roof/wall mounted condensing unit with R32 ment Unit refrigerant rejecting heat from the indoor DX coil. Inverter-driven scroll compressor matched to indoor AHU capacity.	—	75.00kg	£1,900
HVC-PUR-003	Hot-Gas Reheat Coil with EEV — Hot-gas HVAC & CO, Enrich-buy reheat coil with electronic expansion valve ment Unit enabling independent humidity control by reheating dehumidified air without sensible cooling penalty.	—	18.00kg	£720
HVC-PUR-004	G4/F7 Filter Set — Two-stage filter set with HVAC & CO, Enrich-buy G4 pre-filter and F7 fine filter (ePM1 50%) ment Unit installed in the AHU return air path to protect coils and maintain clean room air quality.	—	2.50kg	£65
HVC-PUR-005	In-Duct UV-C Lamp — In-duct germicidal HVAC & CO, Enrich-buy UV-C lamp module sterilising recirculated airment Unit and inhibiting microbial growth on coil surfaces. Includes ballast and safety interlock.	—	3.20kg	£280
HVC-PUR-006	Liquid CO2 Cylinder — Liquid CO2 cylinder HVAC & CO, Enrich-buy supplying enrichment gas to the grow room ment Unit via solenoid-controlled regulator to maintain elevated photosynthesis CO2 setpoint (~1200 ppm).	—	90.00kg	£180
HVC-PUR-007	CO2 Two-Stage Regulator and Solenoid — HVAC & CO, Enrich-buy Two-stage CO2 regulator with integrated 24Vment Unit solenoid shutoff valve for controlled enrichment dosing from bulk CO2 cylinder. Includes high/low pressure gauges and CGA-320 inlet connection.	—	1.80kg	£145
HVC-PUR-008	NDIR CO2 Sensor — Non-dispersive in- HVAC & CO, Enrich-buy frared CO2 sensor module with 0-5000 ppm ment Unit range and 4-20mA output for closed-loop enrichment control. Wall-mount IP54 housing with display.	—	0.25kg	£220

HVC-PUR-009	Temperature/RH Sensor — Combined temperature and relative humidity sensor with Modbus RTU output for grow-room climate monitoring. Operating range -20 to +60°C, 0-100% RH.	HVAC & CO, Enrich-buyment Unit	—	0.12kg	£85
LED-001	LED Dimming Wiring Loom — Pre-assembled wiring loom connecting LED drivers to LED bars and the DALI dimming bus. Includes power conductors, DALI control pair, strain reliefs, and labelled connectors per fixture position.	LED Grow Light System	make	other · Copper conductor with PVC insulation · ±±10mm length	4.50kg £95
LED-002	LED Suspension Bracket — Sheet metal bracket suspending LED light bars from the underside of each grow rack tier. Formed Z-profile with mounting slots for fixture rails and through-holes for M5 fasteners into the rack frame.	LED Grow Light System	make	sheet_metal · Mild Steel · ±±0.5mm	0.35kg £7
LED-ASY	LED Grow Light System Assembly — Complete LED grow light system assembly providing full-spectrum photosynthetic illumination to every grow tier on both sides. Includes LED bars, driver power supplies, mounting hardware, and dimming control wiring back to the farm controller.	LED Grow Light System	make	other · Mixed (Aluminium heatsinks, polycarbonate diffusers, electronics) · ±±2mm (assembly)	180.00kg £12,000
LED-PUR-001	Full-Spectrum LED Grow Bar — Full-spectrum horticultural LED grow bar with integrated heatsink and driver, optimised for leafy greens (lettuce, herbs, brassicas). IP65-rated linear fixture with 0-10V dimming input for DLI control by the farm controller.	LED Grow Light System	buy	—	2.20kg £95
LED-PUR-002	Constant-Current LED Driver — Constant-current LED driver providing regulated DC output to a horticultural LED array. Features 0-10V/PWM/DALI dimming input and overcurrent/overvoltage/overtemperature protection.	LED Grow Light System	buy	—	1.40kg £78
LED-PUR-003	DALI Zone Controller Gateway — DALI-2 zone controller / gateway managing addressable LED drivers across multiple lighting zones. Provides Modbus TCP / BACnet uplink to the farm controller for schedule and DLI control.	LED Grow Light System	buy	—	0.45kg £285
NUT-001	PVC-U Distribution Manifold — Manufactured PVC-U distribution manifold with multiple take-offs feeding the rack tier supply lines from the recirculation pump; solvent-welded assembly with threaded zone valve sockets.	Nutrient Reservoir & Recirculation Skid	make	other · PVC-U · ±±1.0mm	3.80kg £85

NUT-002	Skid Frame Box Section — Welded structural skid base for the nutrient reservoir and recirculation pump system, fabricated from rectangular hollow section steel. Provides a rigid, drip-tray-compatible foundation that bolts to the container floor.	Nutrient Reservoir & make Recirculation Skid	sheet_metal · Mild Steel · ±±1.5mm	28.00kg	£240
NUT-ASY	Nutrient Reservoir & Recirculation Skid Assembly — Top-level assembly of the nutrient reservoir and recirculation skid: tank, pump, dosing pumps, manifolds, valves, sensors, and frame integrated as a pre-piped, pre-wired module ready for connection to the rack distribution loop.	Nutrient Reservoir & make Recirculation Skid	other · Mixed (HDPE tank, 304 SS frame, PVC pipework) · ±±2mm assembly	165.00kg	£4,200
NUT-PUR-001	800L HDPE Nutrient Reservoir Tank — 800 L food-grade HDPE nutrient reservoir tank with lid, level sight strip, and threaded inlet/outlet/drain bosses. Houses the bulk recirculating hydroponic solution.	Nutrient Reservoir & buy Recirculation Skid	—	38.00kg	£420
NUT-PUR-002	Vertical Multistage Recirculation Pump — Vertical multistage centrifugal pump providing recirculation flow to all rack tiers via the distribution manifold. Stainless wetted parts, 3-phase 400V motor with VFD-compatible windings.	Nutrient Reservoir & buy Recirculation Skid	—	28.00kg	£685
NUT-PUR-003	Diaphragm Metering Dosing Pump — Solenoid-driven diaphragm metering pump for precise injection of pH adjuster or A/B nutrient concentrate. 4-20mA / pulse control input, PTFE diaphragm, PVDF wet end.	Nutrient Reservoir & buy Recirculation Skid	—	3.20kg	£320
NUT-PUR-004	Industrial pH Probe — Industrial inline pH probe with PT1000 temperature compensation and BNC/VP6 connector for continuous monitoring of recirculating hydroponic nutrient solution.	Nutrient Reservoir & buy Recirculation Skid	—	0.18kg	£95
NUT-PUR-005	Industrial EC Probe — Industrial inline electrical conductivity (EC/TDS) probe for measuring nutrient strength in the recirculation loop, with integrated temperature sensor and 3/4" BSP threaded body.	Nutrient Reservoir & buy Recirculation Skid	—	0.20kg	£110
NUT-PUR-006	Stainless Mesh Return Strainer — Stainless steel mesh strainer fitted on the return line into the nutrient reservoir to capture root debris and particulates before they enter the pumping system.	Nutrient Reservoir & buy Recirculation Skid	—	0.45kg	£38

NUT-PUR-007	Polypropylene Cartridge Filter Housing — Nutrient Reservoir & buy Polypropylene 10" cartridge filter housing for Recirculation Skid inline 5 micron sediment filtration of the nutrient supply line, with bypass valve and pressure relief.	—	1.10kg	£42
NUT-PUR-008	UVC Inline Steriliser Unit — Inline UVC ster-Nutrient Reservoir & buy iliser unit with 55W amalgam lamp and 316 Recirculation Skid stainless chamber for pathogen control of recirculating nutrient solution at up to 4 m³/h.	—	4.50kg	£320
NUT-PUR-009	Solenoid Zone Valve — 24V normally-closedNutrient Reservoir & buy brass solenoid zone valve controlling nutrientRecirculation Skid flow to each rack tier branch from the distribution manifold.	—	0.35kg	£16
NUT-PUR-010	Ultrasonic Reservoir Level Sensor — Nutrient Reservoir & buy Top-mounted ultrasonic non-contact level Recirculation Skid sensor for measuring nutrient reservoir liquid level with 4-20mA output to the farm controller.	—	0.35kg	£125
SHL-002	Stainless Steel Interior Lining Sheet — Insulated Container make Food-grade stainless steel sheet forming hy-Shell gienic interior wall lining over PIR insulation. Seams sealed with food-safe silicone for washdown compliance.	sheet_metal · 304 Stainless Steel · ±±0.5mm	24.00kg	£110
SHL-003	Reinforced Floor Plate — Reinforced check-Insulated Container make er-plate floor section laid over container baseShell to support rack loads, foot traffic, and pallet trolleys. Coated for chemical and water resistance.	sheet_metal · Mild Steel Checker Plate · ±±1.0mm	95.00kg	£320
SHL-004	GRP Drip Tray and Floor Membrane — Insulated Container make Single-piece moulded GRP drip tray and Shell floor membrane covering the entire container floor, with raised perimeter lip to contain nutrient leaks and direct spillage to drain points.	other · Glass Re- inforced Polyester · ±±3mm	110.00kg	£1,450
SHL-ASY	Insulated Container Shell Assembly — Insulated Container make Top-level assembly of the insulated 40ft HC Shell container shell including ISO frame, PIR insulation, hygienic stainless lining, reinforced floor, personnel door, and embedded rack anchors. Forms the structural and thermal envelope housing all vertical farm subsystems.	other · Mixed (steel/PIR/stainless/GRP) · ±±5mm (ISO 668)	4850.00kg	£18,500
SHL-PUR-001	40ft High Cube ISO Container Shell — Stan-Insulated Container buy dard ISO 668 compliant 40ft High Cube Shell shipping container, used (CW grade) as the structural shell. Provides corner castings, corrugated steel walls, and base rails for transport and mounting.	—	3900.00kg	£4,500

SHL-PUR-002	PIR Sandwich Insulation Panel — Polyiso-cyanurate (PIR) foam sandwich panel with pre-coated steel facings, providing thermal insulation (UH0.22 W/m²K) for walls and ceiling. Cut to size and bonded to interior of container shell.	Insulated Container Shell	buy	—	12.00kg	£95
SHL-PUR-003	Insulated Personnel Door — Insulated personnel access door with food-grade hygienic finish, gasket seal, and lever handle with internal escape mechanism. Mounted in end wall of container.	Insulated Container Shell	buy	—	45.00kg	£650
SHL-PUR-004	M12 Stainless Rack Anchor Insert — M12 stainless steel threaded anchor insert chemically bonded into container floor and wall studs to provide secure attachment points for grow racks and equipment skids.	Insulated Container Shell	buy	—	0.05kg	£4

5. Cost waterfall

Cost estimates grounded against the material-properties library (47 rows, freshest 23 Mar 2026) and the `process\_capabilities` table (20 rows, freshest unknown). Costs sit in the project's ai_cost_estimates jsonb, keyed by module id.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£69,555	£150,000	£80,445

Per-module roll-up

MODULE	COST	% OF UNIT
Insulated Container Shell	£7,200	10.4%
Port-Side Hydroponic Grow Racks	£2,316	3.3%
Starboard-Side Hydroponic Grow Racks	£2,460	3.5%
LED Grow Light System	£26,904	38.7%
Nutrient Reservoir & Recirculation Skid	£13,350	19.2%
HVAC & CO ₂ Enrichment Unit	£12,850	18.5%
Electrical Distribution Panel	£1,405	2.0%
IoT Sensor Mesh & Farm Controller	£3,070	4.4%

6. Risks register

Every failure mode and open question declared against each module, in one register. Until a dedicated risks store ships with severity + ownership, these rows are the canonical risk inventory.

Insulated Container Shell

KNOWN FAILURE MODES (4)

- Condensation and mould growth at thermal bridges where insulation meets corrugations or door frame, contaminating the food-grade environment
- Floor deflection or anchor pull-out under fully-watered rack loads (~6 tonnes distributed) leading to misalignment of nutrient drainage gradients
- Gasket and silicone seam degradation under repeated wash-down cycles causing water ingress into insulation core and loss of R-value
- Corten shell corrosion at floor-to-wall junction where standing nutrient solution contacts unprotected steel

OPEN QUESTIONS (3)

- Confirmed fully-laden operational weight including reservoirs (source 5,000 kg figure appears low; likely 8–10 tonnes) and resulting floor reinforcement spec
- Whether a separate insulated personnel door or modified rear cargo doors better serves weekly planting workflow
- Required fire rating of insulation core (PIR vs mineral wool) under UK Building Regs Approved Doc B for an occupied food-production container

Port-Side Hydroponic Grow Racks

KNOWN FAILURE MODES (4)

- Tray drainage blockage from root mass or biofilm causing flooding, anaerobic conditions, and crop loss
- Frame deflection or racking under wet-tray load (>15 kg/tray fully planted) leading to misaligned LED-to-canopy distances and uneven growth
- Galvanic corrosion at aluminium/stainless interfaces in the high-humidity environment, compromising structural integrity over time
- Biofilm and algal growth on tray surfaces creating food-safety non-conformance and pathogen reservoirs

OPEN QUESTIONS (3)

- Final choice between NFT and shallow DWC tray architecture — affects tray depth, water volume per tier, and reservoir sizing
- Required floor reinforcement detail beneath rack uprights given container baseline floor rating of ~0.5 kN/m²
- Aisle clearance specification (600 vs 800 mm) and its impact on rack depth and tray length tolerance

Starboard-Side Hydroponic Grow Racks

KNOWN FAILURE MODES (4)

- Tray slope drift causing flooding or dry spots, leading to root death or algal contamination
- Galvanic or chloride corrosion at steel/aluminium/stainless interfaces in persistent high-RH/nutrient-mist environment
- Floor anchor pull-out or rack sway under transport vibration during flatbed delivery
- Biofilm build-up in return gutters causing cross-tier disease transfer

OPEN QUESTIONS (3)

- Final tier pitch trade-off between LED-to-canopy distance and total tier count to hit exactly 3,000 slots
- Whether NFT or shallow DWC is chosen as the hydroponic technique (affects tray depth and return flow rate)
- Aisle clearance dimension once port and starboard racks are both installed (target 700 mm minimum)

LED Grow Light System

KNOWN FAILURE MODES (4)

- Driver failure from humidity ingress or thermal cycling, taking out an entire tier zone until replaced
- Gradual photon flux degradation (LM-80/L90 decay) reducing DLI below crop setpoint and silently cutting yield if not compensated by the controller
- Corrosion of connectors and mains drops in the high-RH, nutrient-mist environment leading to earth faults and RCD trips
- Hotspot formation or non-uniform PPFD distribution at canopy edges causing uneven crop growth and harvest losses

OPEN QUESTIONS (3)

- Final bar count, wattage, and spectrum recipe per crop (lettuce vs herbs vs brassicas) needed to hit target DLI within the lighting energy sub-budget
- Whether 0–10 V analogue dimming is sufficient or full DALI-2 addressable control is required for per-tier tuning and fault reporting
- IP rating margin and conformal coating strategy required for long-term reliability at 70–80 % RH with periodic wash-down

Nutrient Reservoir & Recirculation Skid

KNOWN FAILURE MODES (4)

- Root-zone pathogen outbreak (Pythium, Fusarium) due to UV lamp degradation or filter bypass, contaminating the entire recirculating loop
- Dosing pump runaway or stuck check-valve causing pH crash or EC overshoot, killing crops before alarm thresholds trigger
- Pump cavitation or blockage from accumulated biofilm/root debris in return strainer, starving upper tiers of feed
- Reservoir or pipework leak onto container floor, leading to electrical hazard and structural corrosion of the steel container shell

OPEN QUESTIONS (3)

- Whether NFT, DWC, or ebb-and-flood is the chosen rack hydraulic regime — this changes flow rate, head, and reservoir sizing materially
- UK water authority requirements for backflow protection on mains top-up (WRAS Fluid Category 5 likely mandates an AB air gap rather than RPZ)
- Disposal/refresh strategy for spent nutrient solution — full dump cycle frequency drives effluent handling and water consumption

HVAC & CO₂ Enrichment Unit

KNOWN FAILURE MODES (4)

- Condensate drain blockage causing water ingress into electrical panel and biofilm/Legionella growth in pan
- Refrigerant leak from corroded coil in high-humidity wash-down environment, leading to loss of cooling and crop loss within 12 hours
- CO₂ regulator or solenoid failure in the open position, creating an asphyxiation hazard for service personnel entering the container
- Filter loading not detected, causing airflow drop, hotspots in canopy, and elevated humidity triggering fungal disease outbreak

OPEN QUESTIONS (3)

- Peak simultaneous sensible + latent load envelope under full lighting at lights-on transition — needs CFD or psychrometric modelling against confirmed transpiration rate
- Whether a single split system meets redundancy expectations for unattended weekly operation or if N+1 modular units are required within budget
- UK F-gas regulation impact on refrigerant choice (R454B vs R32) and certified installer requirements for serial production

Electrical Distribution Panel

KNOWN FAILURE MODES (4)

- Nuisance RCD trip from cumulative LED driver leakage current causing total crop loss within hours
- Contactor weld-on under inrush from large LED driver banks, preventing scheduled lights-off and overheating the grow zone
- Condensation ingress in humid grow environment causing corrosion of busbars and terminal blocks
- Phase imbalance from poorly distributed single-phase loads tripping the incomer or overheating one phase conductor

OPEN QUESTIONS (3)

- Final per-phase load schedule and whether 63A or 100A incomer is required once LED driver efficiency is fixed
- Whether site supply is TN-S, TN-C-S or TT — affects earthing arrangement and RCD selection
- Need for an integrated UPS or generator changeover to ride through short outages and protect the nutrient pumps and controller

IoT Sensor Mesh & Farm Controller

KNOWN FAILURE MODES (4)

- pH/EC probe drift or fouling causing incorrect dosing and crop damage if not auto-calibrated or flagged
- Loss of 4G/Ethernet uplink leaving the unit operating blind to the remote dashboard; mitigated by local autonomous control but loses visibility
- Wireless mesh interference or node battery exhaustion causing stale telemetry and missed alarms
- Controller firmware lockup or SD-card corruption halting all setpoint commands; requires watchdog and redundant boot media

OPEN QUESTIONS (3)

- Choice between fully wired Modbus/RS-485 sensor bus versus wireless LoRaWAN/Zigbee mesh — trade-off of installation cost vs reliability in a metal container RF environment
- Cybersecurity and OTA update strategy for fleet of remotely-managed units (certificate management, VPN vs MQTT broker)
- Required sensor redundancy level for pH/EC — single probe with calibration schedule or dual-probe voting for fail-safe dosing

7. Supplier shortlist (32)

Shortlist built by scoring each supplier in the directory (651 companies) and marketplace listings (28,315 listings) against each module's declared process + material. A low shortlist count usually means the directory doesn't yet have coverage for the project's niche — not that no match exists globally.

accupanels.com

DC DISTRIBUTION BOARD

<https://www.accupanels.com/dc-distribution-board/>

HQ not declared · ramp role unassigned · match score 30.5

Matched across 1 module: Electrical Distribution Panel

SCORE BREAKDOWN

total	30.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.1
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

industratech.uk

Industratech

<https://industratech.uk/control-panel-manufacturers/>

Contact: sales@industratech.co.uk

HQ not declared · ramp role unassigned · match score 29.2

Matched across 1 module: Electrical Distribution Panel

SCORE BREAKDOWN

total	29.2
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.8
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Products

airgreeninc.com

Indoor Agriculture Climate Control

<https://airgreeninc.com/industries-served-applications/indoor-agriculture/>

HQ not declared · ramp role unassigned · match score 29.3

Matched across 1 module: hvac_unit

SCORE BREAKDOWN

total	29.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.9
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

airproducts.com

CO2 Gas Supplier

<https://www.airproducts.com/gases/carbon-dioxide>

HQ not declared · ramp role unassigned · match score 26.7

Matched across 1 module: co2_dosing

SCORE BREAKDOWN

total	26.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	26.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

eleics.com

Eleics Design Private Limited

<https://www.eleics.com/energy-management-and-monitoring>

Contact: info@eleics.com

HQ not declared · ramp role unassigned · match score 28.2

Matched across 1 module: IoT Sensor Mesh & Farm Controller

SCORE BREAKDOWN

total	28.2
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	24.8
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Products

keswitchgear.co.uk

G99 Protection Panels

<https://www.keswitchgear.co.uk/renewables/g99-protection-panels/>

Contact: info@keswitchgear.co.uk

HQ not declared · ramp role unassigned · match score 29.3

Matched across 1 module: Electrical Distribution Panel

SCORE BREAKDOWN

total	29.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

sunplusled.com

Custom Multi Tier Indoor Cultivation Plant Led Grow System Movable Rolling Hydroponic Vertical Farming Mobile Grow Rack

<https://sunplusled.com/products/custom-multi-tier-indoor-cultivation-plant-led-grow-system-movable-rolling-hydroponic-vertical-farming-mobile-grow-rack>

Contact: info@sunplusled.com

HQ not declared · ramp role unassigned · match score 31.0

Matched across 1 module: Starboard-Side Hydroponic Grow Racks

SCORE BREAKDOWN

total	31
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	30.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

montel.com

Vertical Grow Rack Systems

<https://www.montel.com/en/solutions/vertical-grow-rack-systems>

Contact: system@montel.com

HQ not declared · ramp role unassigned · match score 29.5

Matched across 1 module: Starboard-Side Hydroponic Grow Racks

SCORE BREAKDOWN

total	29.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	29.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

lightera.net

Interlight 2.8umol/J PPE full spectrum LED grow Light fixture for indoor & Greenhouse Solution

<https://www.lightera.net/products/led-grow-light/interlight/Horticulture/886.html>

Contact: sales@sanan-e.com

HQ not declared · ramp role unassigned · match score 31.3

Matched across 1 module: LED Grow Light System

SCORE BREAKDOWN

total	31.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

phlizon.co.uk

LED Grow Lights, full-spectrum grow light

<https://www.phlizon.co.uk/>

Contact: support@phlizonstore.com

HQ not declared · ramp role unassigned · match score 30.5

Matched across 1 module: LED Grow Light System

SCORE BREAKDOWN

total	30.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	30.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

pattersonpope.com

Mobile GROW Rack

<https://pattersonpope.com/products/high-density-storage-systems/mobile-grow-rack/>

HQ not declared · ramp role unassigned · match score 28.9

Matched across 1 module: Starboard-Side Hydroponic Grow Racks

SCORE BREAKDOWN

total	28.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.5
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

cfstruthers.co.uk

CFS

<https://www.cfstruthers.co.uk/services-and-installation>

Contact: sales@cfstruthers.co.uk

HQ not declared · ramp role unassigned · match score 29.3

Matched across 1 module: Insulated Container Shell

SCORE BREAKDOWN

total	29.3
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	18.9
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Marine
- Heavy Fabrication & Pressure Vessels

mobilemodularcontainers.com

Hydroponic Shipping Container Farms: 4,000-8,000 Plants in 320 Sq Ft, from \$15K DIY to \$180K Turnkey

<https://www.mobilemodularcontainers.com/blog/hydroponics-in-shipping-containers>

Contact: privacyinquiry@mgrc.com

HQ not declared · ramp role unassigned · match score 27.5

Matched across 2 modules: Insulated Container Shell, grow_racks_left

SCORE BREAKDOWN

total	27.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

ledinside.com

Atop HortiBar LED Interlighting Grow Light for Horticulture Lighting

https://www.ledinside.com/press/2017/7/atop_hortibar_led_interlighting_grow_light_for_horticulture_lighting

HQ not declared · ramp role unassigned · match score 29.9

Matched across 1 module: LED Grow Light System

SCORE BREAKDOWN

total	29.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	29.5
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

atlas-scientific.com

The Ultimate Guide To Nutrient Dosing Systems

<https://atlas-scientific.com/blog/nutrient-dosing-systems/>

Contact: info@atlas-scientific.com

HQ not declared · ramp role unassigned · match score 26.5

Matched across 1 module: Nutrient Reservoir & Recirculation Skid

SCORE BREAKDOWN

total	26.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	26.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

priva.com

Indoor farming integrated climate control systems

<https://www.priva.com/indoor-growing/integrated-climate-control>

Contact: yasmin.brand@priva.com

HQ not declared · ramp role unassigned · match score 26.5

Matched across 1 module: `iot_control`

SCORE BREAKDOWN

total	26.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	26.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

isoclad.co.uk

Isoclad

<https://www.isoclad.co.uk/company-overview/>

Contact: info@isoclad.co.uk

HQ not declared · ramp role unassigned · match score 26.7

Matched across 1 module: Insulated Container Shell

SCORE BREAKDOWN

total	26.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Insulated Panels & Cladding Systems

sensorsandtransmitters.com

Co2 Fertilisation

<https://sensorsandtransmitters.com/co2-fertilisation-guide-to-monitoring-carbon-dioxide-in-greenhouses/>

HQ not declared · ramp role unassigned · match score 30.7

Matched across 1 module: co2_dosing

SCORE BREAKDOWN

total	30.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

soshygiene.com

Dosatron D3RE Water Powered Chemical Dosing Pump – Blue or PVDF, Multiple Injection Options

<https://soshygiene.com/products/dosatron-d3re-dosing-pump-chemical-injector-unit>

HQ not declared · ramp role unassigned · match score 26.4

Matched across 1 module: Nutrient Reservoir & Recirculation Skid

SCORE BREAKDOWN

total	26.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	26
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

artisuntechnology.com

Mixing Guide for Nutrient Solution for Hydroponics System RDWC System

<https://www.artisuntechnology.com/blogs/our-farm/mixing-guide-for-nutrient-solution-for-hydroponics-system-rdwc>

HQ not declared · ramp role unassigned · match score 26.0

Matched across 1 module: Nutrient Reservoir & Recirculation Skid

SCORE BREAKDOWN

total	26
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

wehydroponics.com

NFT / DFT Channels 100*80mm

<https://www.wehydroponics.com/store/product/nft-channels-80mm-3-meter/>

Contact: indore@wehydroponics.com

HQ not declared · ramp role unassigned · match score 27.6

Matched across 2 modules: grow_racks_left, grow_racks_right

SCORE BREAKDOWN

total	27.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

grow-tec.com

HVAC

<https://grow-tec.com/hvac/>

Contact:

HQ not declared · ramp role unassigned · match score 30.8

Matched across 1 module: HVAC & CO, Enrichment Unit

SCORE BREAKDOWN

total	30.8
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	30.4
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

specialist-coatings.co.uk

Specialist Coatings (GB) Limited

<https://www.specialist-coatings.co.uk/>

Contact: info@specialist-coatings.co.uk

HQ not declared · ramp role unassigned · match score 29.6

Matched across 1 module: Insulated Container Shell

SCORE BREAKDOWN

total	29.6
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	19.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Marine
- Services

sltmaks.com

Vertical Farming LED Lighting: Space, Heat, and Yield Guide

<https://www.sltmaks.com/vertical-farming-led-lighting.html>

HQ not declared · ramp role unassigned · match score 29.6

Matched across 1 module: led_lighting

SCORE BREAKDOWN

total	29.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	29.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

hydro-harvest.co.uk

Pro-Leaf Co2 Controller

<https://www.hydro-harvest.co.uk/product/pro-leaf-co2-controller/>

Contact:

HQ not declared · ramp role unassigned · match score 28.7

Matched across 1 module: co2_dosing

SCORE BREAKDOWN

total	28.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

dosingpump.shop

Dosing systems for horticulture

<https://dosingpump.shop/en/products/solutions-completes/horticulture-and-agriculture/>

Contact: info@dosingpump.shop

HQ not declared · ramp role unassigned · match score 25.6

Matched across 1 module: Nutrient Reservoir & Recirculation Skid

SCORE BREAKDOWN

total	25.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

prominent.com

ProMinent

<https://www.prominent.com/en/Products/Products/Membrane-Technology-and-Membrane-Filtration/pg-membrane-technology-and-filtration.html>

HQ not declared · ramp role unassigned · match score 26.8

Matched across 1 module: Nutrient Reservoir & Recirculation Skid

SCORE BREAKDOWN

total	26.8
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.4
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Membrane Filtration & Metering Systems

dfrobot.com

Unleash Precision with RS485 Soil EC and pH Sensor

<https://www.dfrobot.com/product-2829.html>

HQ not declared · ramp role unassigned · match score 28.9

Matched across 1 module: IoT Sensor Mesh & Farm Controller

SCORE BREAKDOWN

total	28.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

cultivasystems.com

Indoor Cannabis HVAC From Cultiva Systems

<https://cultivasystems.com/indoor-cannabis-hvac/>

Contact:

HQ not declared · ramp role unassigned · match score 30.6

Matched across 1 module: HVAC & CO, Enrichment Unit

SCORE BREAKDOWN

total	30.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	30.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

hyper-logic.com

Hydroponic Wastewater & Nutrient Runoff Treatment

<https://hyper-logic.com/waste-water-solutions-for-agriculture/>

HQ not declared · ramp role unassigned · match score 27.4

Matched across 1 module: Nutrient Reservoir & Recirculation Skid

SCORE BREAKDOWN

total	27.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

altaqua.com

Cannabis Grow Room HVAC

<https://altaqua.com/cannabis-grow-room-hvac>

Contact:

HQ not declared · ramp role unassigned · match score 30.0

Matched across 1 module: HVAC & CO, Enrichment Unit

SCORE BREAKDOWN

total	30
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	29.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

uwyo.edu

Argus Titan 2 System

<https://www.uwyo.edu/science-initiative/plant-facility/argus-system.html>

HQ not declared · ramp role unassigned · match score 26.4

Matched across 1 module: IoT Sensor Mesh & Farm Controller

SCORE BREAKDOWN

total	26.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	26
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing